Team Details

- ➔ Team name - **SuperPumped**
- ➔ Team leader name - **Kush**
- ➔ Problem Statement - **Residential societies lack a proactive, privacy-preserving way to automatically detect unauthorized parking and stray animal threats from 24/7 CCTV feeds without relying on expensive, privacy-invasive cloud AI.**

PROJECT OVERVIEW

ColonyEdge: Privacy-First Edge AI

Secure and localized computer vision for smart city residential safety, powered by AMD Ryzen AI.

Kush,Luv
Presenter

THE CHALLENGE

The Hidden Costs of Cloud Security

Addressing the critical privacy and financial vulnerabilities in modern residential surveillance systems.

01

Privacy Violations: Constant streaming of 24/7 video feeds to cloud servers poses a massive security risk to residents.

02

Crippling Bandwidth: High costs associated with continuous video data transmission over public networks.

03

High Latency: Cloud-dependent processing delays response times for critical safety threats.

A Paradigm Shift to Edge Inference

ColonyEdge pivots to 100% offline edge inference, ensuring no raw video ever leaves the premises.

Localized Edge Intelligence

Optimized models run on AMD Ryzen AI NPUs for millisecond decision-making.

Privacy-First Design

Like an iceberg tip: only lightweight metadata is visible; raw video stays local.

Passive→Active Sensors

Convert CCTV into proactive sensors that alert security before incidents escalate.

CORE FEATURES

Intelligent Security Features

Specialized computer vision modules designed for the unique needs of residential societies.

Stray Animal Detection

Real-time identification of stray dog packs in zones Automated alerts for play areas and basements

Smart Parking Simulation

Unauthorized vehicle detection in resident slots Cross-reference against local flat-allocation databases

Command Center Dashboard

React.js-based dark-mode security interface Instant WebSocket-based threat notifications

The Edge Inference Workflow

How video frames are converted into actionable intelligence in milliseconds.

Input

Society CCTV cameras capture live video frames for local processing.

Edge Processing

Local *AMD Ryzen AI* NPU runs *YOLOv8* inference on incoming frames.

Logic Gate

AI detects threats and checks local parking databases for authorization.

Transmission

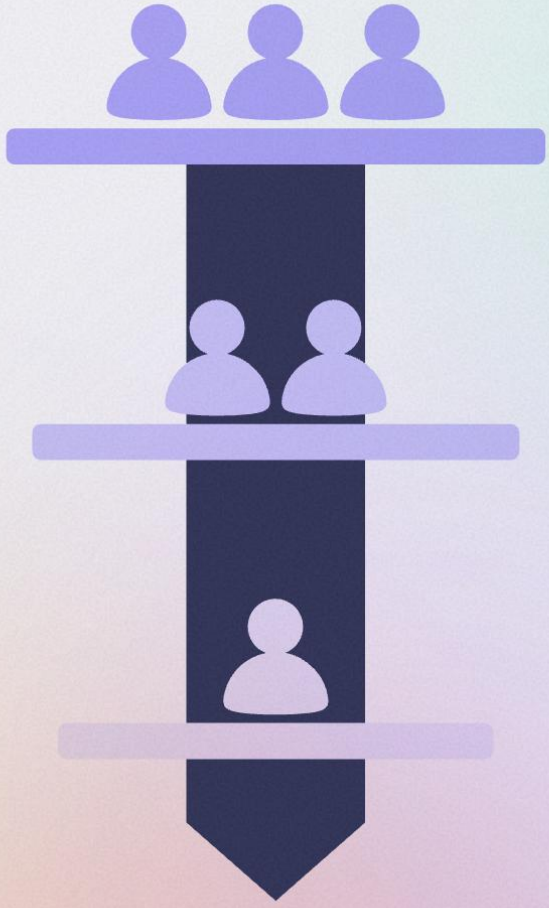
The edge node fires a lightweight JSON payload via *FastAPI* WebSockets.

Output

The *React.js* dashboard flashes visual alerts for security personnel.

System Architecture Overview

A robust three-tier architecture connecting hardware sensing with local monitoring.



Hardware Edge

CCTV feeds routed to local security PCs powered by *AMD Ryzen AI NPUs*.

Software Edge

OpenCV extraction and *YOLOv8* object detection powered by *FastAPI*.

Frontend Interface

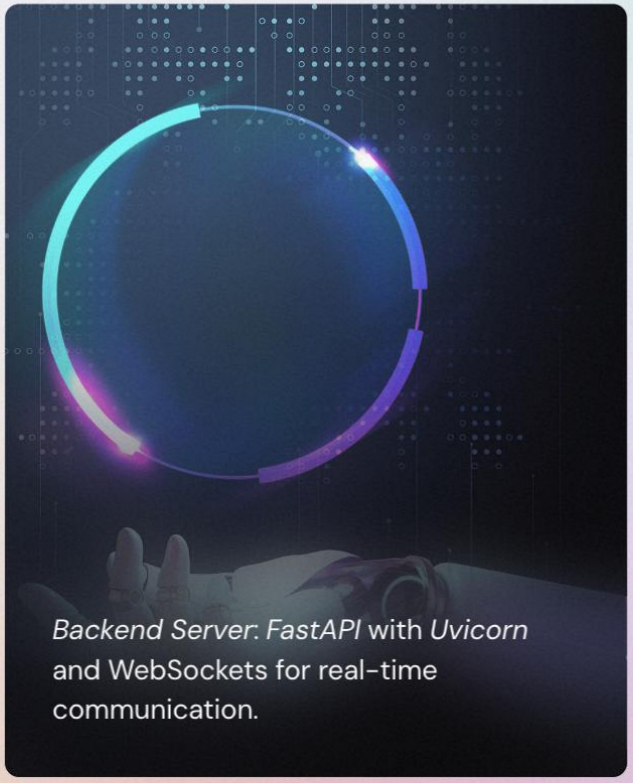
Local Area Network connectivity to the *React.js* Security Dashboard.

TECH STACK

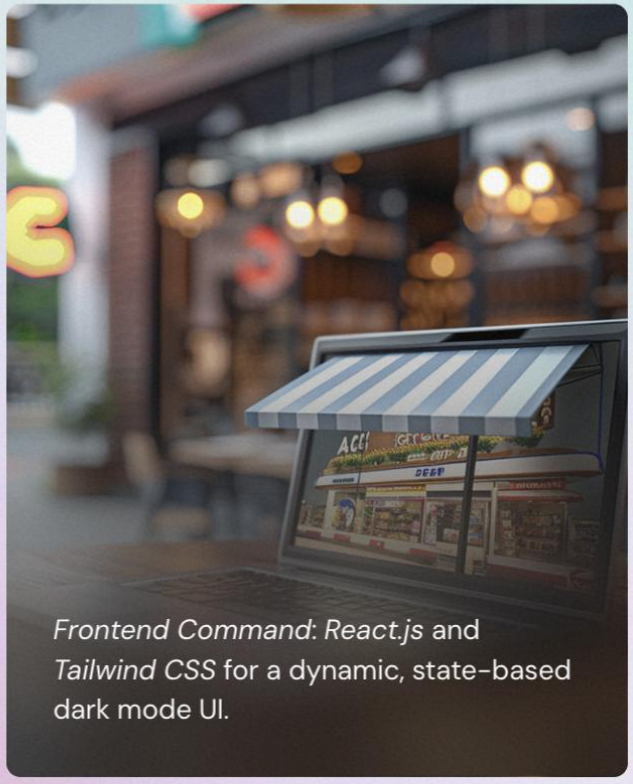
Modern Technology Stack

Leveraging industry-standard tools optimized for edge computing efficiency.

Computer Vision: Ultralytics *YOLOv8 Nano* optimized for Edge deployment with *OpenCV*.



Backend Server: *FastAPI* with *Uvicorn* and WebSockets for real-time communication.



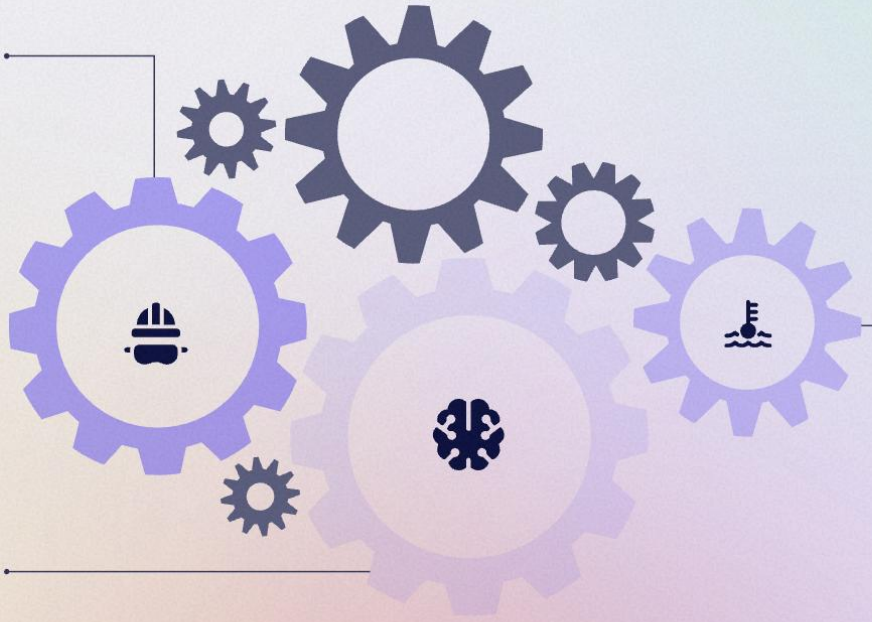
Frontend Command: *React.js* and *Tailwind CSS* for a dynamic, state-based dark mode UI.

Efficiency with AMD Ryzen AI

Offloading 24/7 inference to the NPU for sustainable and cost-effective AI operations.

NPU Offloading

Moving matrix multiplications to the purpose-built *AMD Ryzen AI* NPU to prevent CPU throttling.



Thermal Efficiency

Continuous 24/7 processing at a fraction of a watt, maintaining system longevity.

Intelligent Edge

Utilizing *AMD's* push for decentralized computing to make local AI financially viable.

ECONOMICS

Cost Efficiency & ROI

Comparing traditional cloud-based AI with the decentralized ColonyEdge model.

Feature	Traditional Cloud AI	ColonyEdge (AMD Node)
Recurring Cost	$500+ / Month	$0 (Self-Hosted)
Data Privacy	High Risk (Cloud Sync)	100% Private (Local)
Hardware Type	Any Basic Camera	AMD-powered Edge PC
Investment	Opex Intensive	Capex (One-time)

PROTOTYPE

Prototype & Project Assets

Explore the code and see the live system demonstration.

GitHub Repository

Access the full source code at Kushack07/ColonyEdge-AMD-Slingshot Explore our model optimization and backend implementation

Demo Video

Watch the system detect threats in real-time on YouTube See the React Security Dashboard alerts in action

CONCLUSION

Securing Smart Cities Locally

ColonyEdge: The future of private, localized, and efficient urban safety.

AMD
Slingshot
HUMAN IMAGINATION
BUILT WITH AI
Powered by H2S
Thank you!